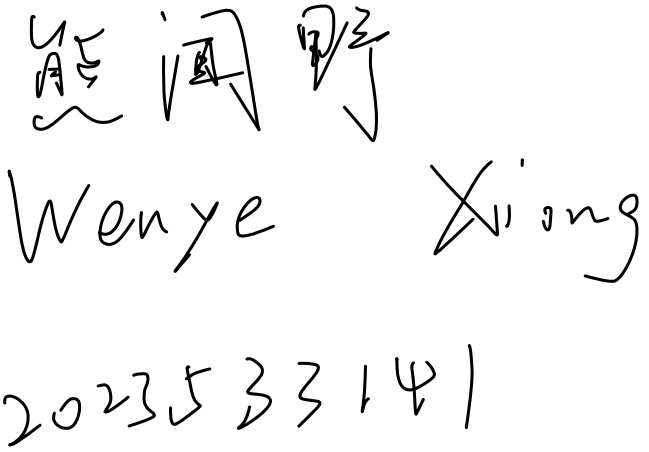
Homework 1

Due: Oct. 8, 11:59 am

Submit your homework outside the instructor's office:
Room 324, Building 3, SIST

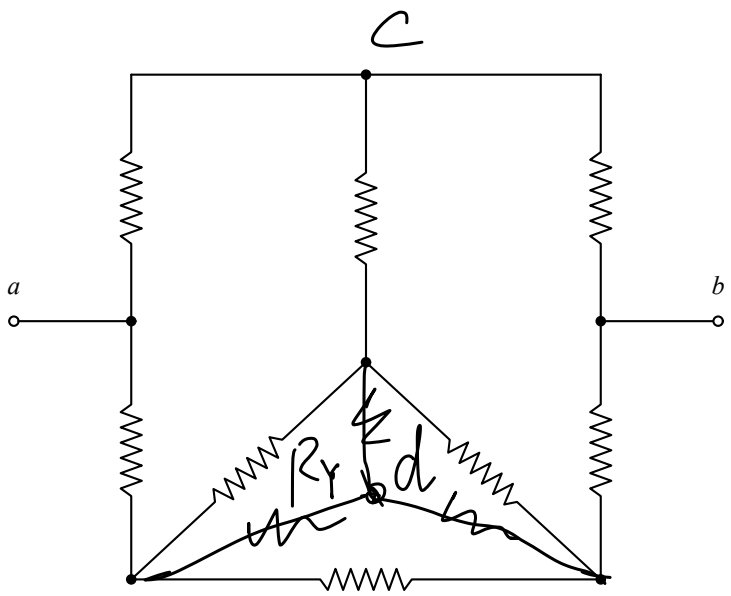
Rules

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism.
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit on time. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain 2 decimal places if rounding is required (unless otherwise specified).



1

Each resistor has a resistance of 10 Ω. Use a Delta ↔ Y transformation to calculate the equivalent resistance R_{ab}.



All resistors: 10 Ω

For a Δ → Y transformation, each new Y is

$$R_Y = \frac{R_\Delta R_\Delta}{R_\Delta + R_\Delta + R_\Delta} = \frac{100}{30} = \frac{10}{3}\ \Omega$$

The new network is a balanced bridge as c, d are at the same potential

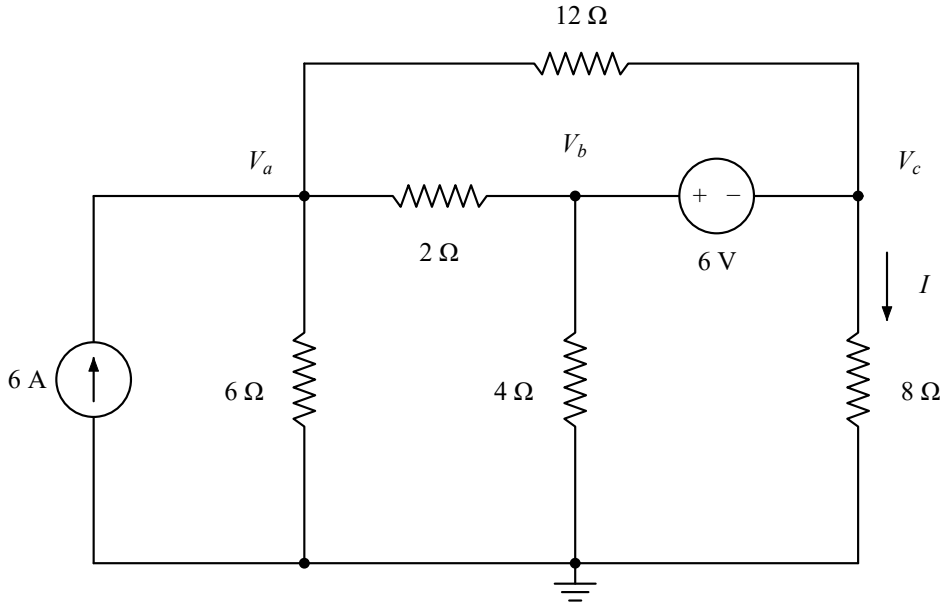
$$R_{ab} = (10+10) \parallel \left(\frac{40}{3} + \frac{40}{3}\right)$$

$$= \frac{20 \cdot \frac{80}{3}}{20 + \frac{80}{3}} = \frac{80}{7}\ \Omega$$

$$= 11.43\ \Omega$$

2

Use nodal analysis to find V_a, V_b, V_c, and I. Find the power absorbed by each element and verify that the total power is zero.



KCL at V_a: $\frac{V_a}{6} + \frac{V_a - V_b}{2} + \frac{V_a - V_c}{12} = 6$ (1)

KCL at V_b, V_c: $\frac{V_b}{4} + \frac{V_b - V_a}{2} + \frac{V_c}{8} + \frac{V_c - V_a}{12} = 0$ (2)

6V Source: $V_b - V_c = 6$ (3)

Solving (1) (2) (3):

$V_a = \frac{1728}{109}\text{V} \approx 15.85\text{ V}$ $V_b = \frac{1194}{109}\text{V} \approx 10.95\text{ V}$

$V_c = \frac{540}{109}\text{V} \approx 4.95\text{ V}$ $I = \frac{V_c}{8} = \frac{135}{218}\text{A} \approx 0.62\text{A}$

$P_{6V} = 6 \cdot I_{6V} = 6 \times (-\frac{65}{218}) \approx -1.73\text{ W}$

Power:

$P_{6A} = -I \cdot V_a = -6 \cdot \frac{1728}{109} \approx -95.12\text{W}$

$P_{6\Omega} = \frac{V_a^2}{6} = \frac{(1728/109)^2}{6} \approx 41.89\text{ W}$

$P_{2\Omega} = \frac{(V_a - V_b)^2}{2} = \frac{(534/109)^2}{2} \approx 12.00\text{ W}$

$P_{12\Omega} = \frac{(V_a - V_c)^2}{12} = \frac{(1188/109)^2}{12} \approx 9.90\text{ W}$

$P_{4\Omega} = \frac{V_b^2}{4} = \frac{(1194/109)^2}{4} \approx 30.00\text{ W}$

$P_{8\Omega} = \frac{V_c^2}{8} = \frac{(540/109)^2}{8} \approx 3.07\text{ W}$

$P_{sources} = P_{6A} + P_{6V} \approx -96.85\text{ W}$

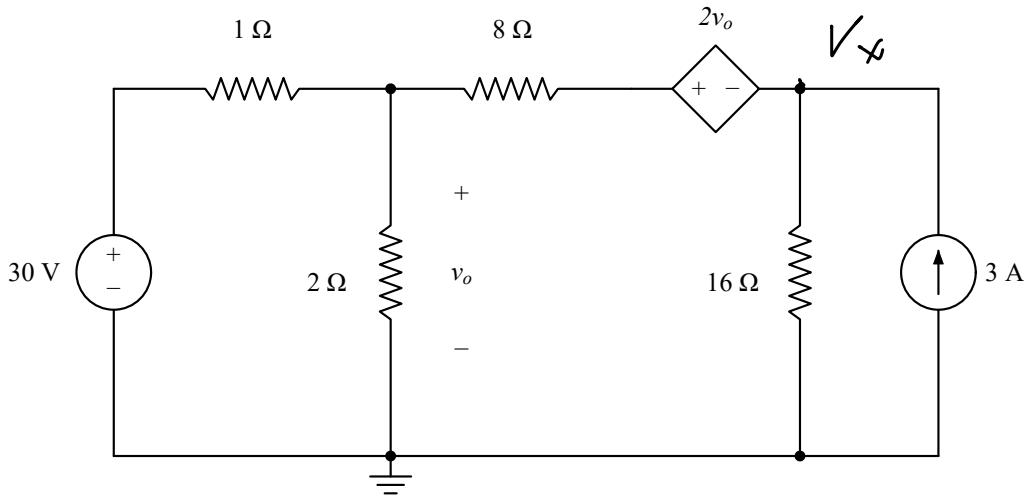
$P_{resistors} = P_{6\Omega} + P_{2\Omega} + P_{12\Omega} + P_{4\Omega} + P_{8\Omega}$

$= 96.85\text{ W}$

$P_{total} = P_{sources} + P_{resistors} = 0\text{ W}$

3

Use nodal analysis to find v_o in the circuit below.



$i_{8\Omega} = \dfrac{v_o - (V_x + 2v_o)}{8} = \dfrac{-v_o - v_x}{8}$

At node v_o:

$$\frac{v_o - 30}{1} + \frac{v_o}{2} + \frac{v_o - (v_x + 2v_o)}{8} = 0$$

$$\Rightarrow 11 v_o - 240 = v_x \quad (1)$$

At node v_x:

$$\frac{(v_x + 2v_o) - v_o}{8} + \frac{v_x}{16} - 3 = 0$$

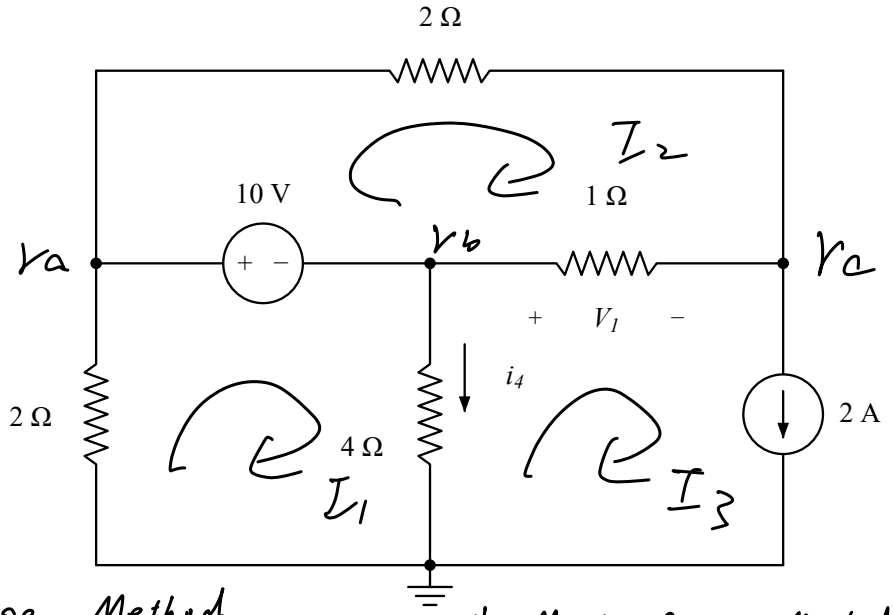
$$\Rightarrow 2v_o + 3v_x = 48 \quad (2)$$

Solving (1) and (2):

$$v_o = \frac{768}{35} \approx 21.94 \text{ V}$$

4

Calculate the branch current through the 4 Ω resistor and the voltage V_1 in the circuit below using (a) the node-voltage method and (b) the mesh-current method.



(a) Node-Voltage Method

$V_a = V_b + 10$ (1)

KCL at Supernode (V_a, V_b):

$$\frac{V_a}{2} + \frac{V_a - V_c}{2} + \frac{V_b}{4} + \frac{V_b - V_c}{1} = 0 \quad (2)$$

KCL at Node V_c:

$$\frac{V_c - V_a}{2} + \frac{V_c - V_b}{1} + 2 = 0 \quad (3)$$

Solving (1) (2) (3):

$V_b = -\frac{28}{3}\,V \approx -9.33\,V$

$V_c = -\frac{28}{3} + 2 = -\frac{22}{3}\,V \approx -7.33\,V$

$i_4 = \frac{V_b}{4} = -\frac{7}{3}\,A \approx -2.33\,A$

$V_1 = V_b - V_c = -2\,V$

(b) Mesh-Current Method

Three Meshes I_1, I_2, I_3 as shown:

$I_2 = 2A$

$i_4 = I_1 - I_3 = I_1 - 2$

$I_{1\Omega} = I_2 - I_3 = I_2 - 2 \Rightarrow V_1 = 1\Omega \cdot I_{1\Omega} = I_2 - 2$

$I_2 - I_1 = I_{\text{branch}}$

KVL for Mesh 1:

$2I_1 + 10 + 4(I_1 - I_3) = 0$

$\Rightarrow 6I_1 - 4I_3 = -10$

Mesh 3: $I_3 = 2A$

Mesh 2: $2I_2 + 1(I_2 - I_3) - 10 = 0$

$\Rightarrow 3I_2 - I_3 = 10$

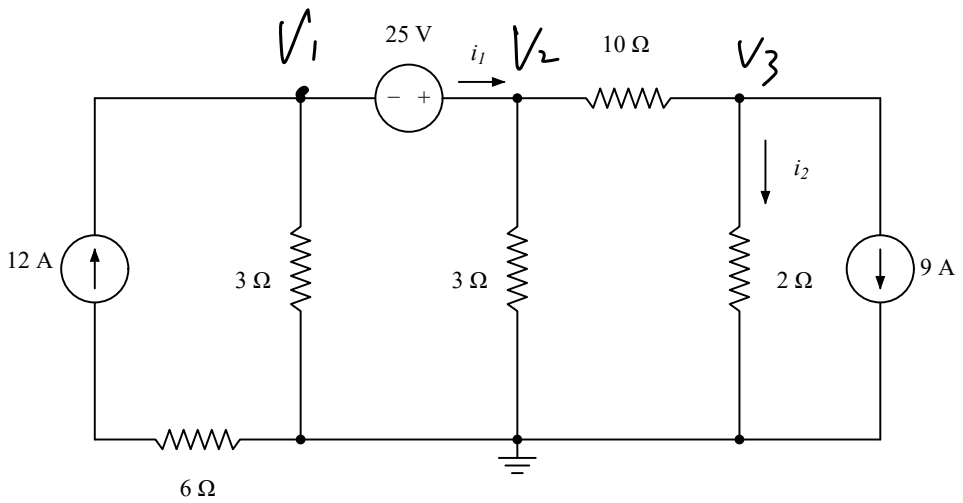
$$\Rightarrow \begin{cases} I_1 = -\frac{1}{3}\,A \\ I_2 = 4\,A \\ I_3 = 2\,A \end{cases}$$

$i_4 = I_1 - I_3 = -2.33\,A$

$V_1 = 1\Omega \cdot (I_3 - I_2) = -2\,V$

5

Use either nodal analysis or mesh analysis to find i_1 and i_2.



Node Analysis:

KCL at Node V_3:

$$\frac{V_3-V_2}{10}+\frac{V_3}{2}+9=0 \quad (1)$$

Node V_1:

$$-12+\frac{V_1}{3}+i_1=0 \quad (2)$$

Node V_2:

$$-i_1+\frac{V_2}{3}+\frac{V_2-V_3}{10}=0 \quad (3)$$

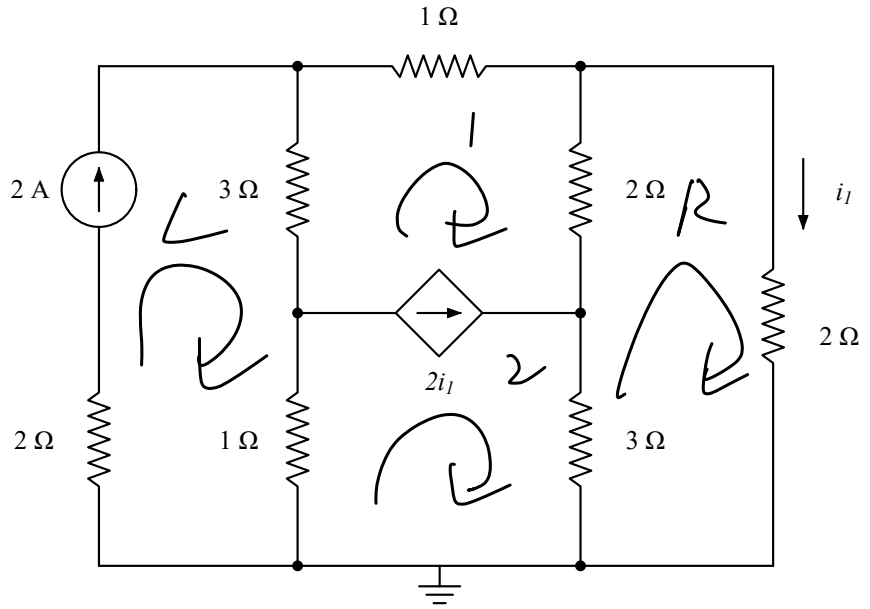
Also, $V_2-V_1=25\,V$ (4)

Solving (1) (2) (3) (4):

$$\begin{cases} V_1=\frac{1}{9}\,V \approx 0.11\,V \\ V_2=\frac{226}{9}\,V \approx 25.11\,V \\ V_3=-\frac{292}{27}\,V \approx -10.81\,V \end{cases} \Rightarrow \begin{cases} i_1=\frac{323}{27}\,A \approx 11.96\,A \\ i_2=\frac{V_3}{2}=-\frac{146}{27}\,A \approx -5.41\,A \end{cases}$$

6

Use the mesh current analysis method to find i_1.



KVL for Mesh 3:

$2(I_3 - I_1) + 3(I_3 - I_2) + 2I_{12} = 0 \quad (1)$

Mesh (1, 2):

$1 \cdot (I_2 - I_L) + 3(I_1 - I_L) + 1 I_1 + 2(I_1 - I_3) + 3(I_2 - I_3) = 0$

$I_L = 2A$

$\Rightarrow 6 I_1 + 4 I_2 - 5 I_3 = 8 \quad (2)$

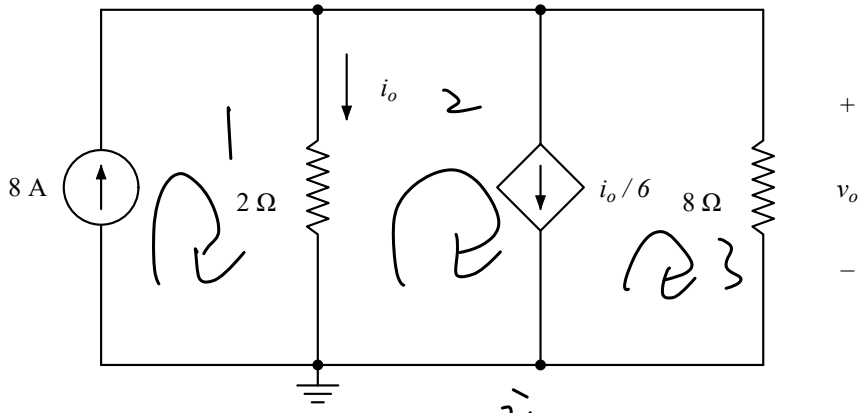
Solving (1), (2) $\Rightarrow I_2 = I_1 + 2I_{12} \quad I_{12} = 5 I_1$

$I_2 = 11 I_1 \Rightarrow 25 I_1 = 8 \Rightarrow I_1 = \frac{8}{25} = 0.32A$

$i_1 = I_3 - 5 I_1 = \frac{8}{5} A = 1.6A$

7

Use the mesh current analysis method to find v_o and i_o in the circuit below.



$i_0 = I_1 - I_2 = 8 - I_2$ $\quad I_2 - I_3 = \frac{i_0}{6}$ $\quad v_0 = 8I_3$

$I_2 - I_3 = \frac{8 - I_2}{6}$ $\quad I_3 = \frac{7I_2 - 8}{6}$

Mesh 2+3: $\quad 2(I_2 - I_1) + 8I_3 = 0$, $I_1 = 8A$

$\Rightarrow I_2 + 4I_3 = 8$

$\Rightarrow I_2 = \frac{40}{17} A \approx 2.35A$

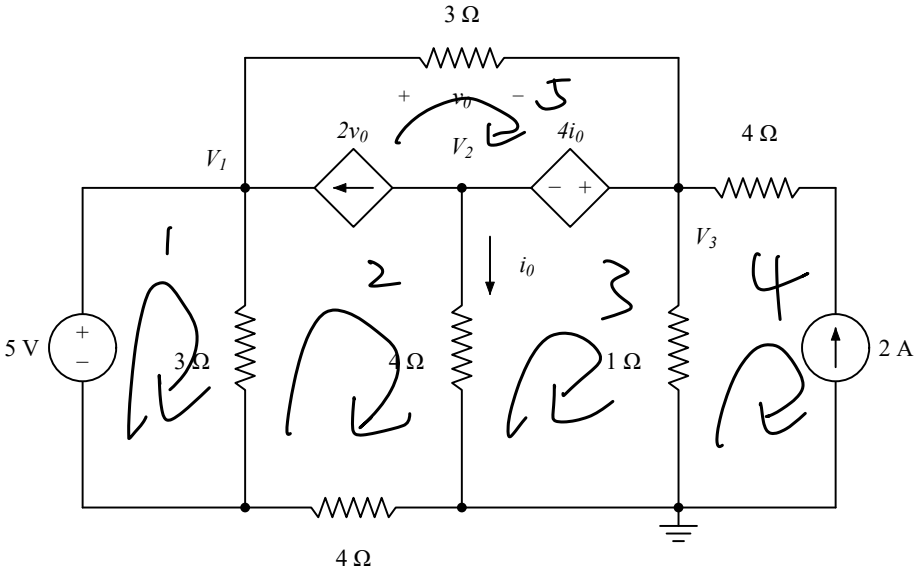
$I_3 = \frac{24}{17} A \approx 1.41 A$

$i_0 = 8 - I_2 = \frac{96}{17} A \approx 5.65A$

$v_0 = 8I_3 = \frac{192}{17} V \approx 11.29 V$

8

Use the mesh current analysis method to find the node voltages V_1, V_2, and V_3 for the circuit below, relative to the indicated ground.



$I_4 = -2A$

$v_0 = 3I_5$

$i_0 = I_2 - I_3$

$I_5 - I_2 = 2v_0 = 6I_5$

$\Rightarrow I_5 = -\frac{1}{5}I_2$

KVL for Mesh 1:

$-5 + 3(I_1 - I_2) = 0$

$\Rightarrow 3(I_2 - I_1) = -5V$

KVL for Mesh 3:

$-4i_0 - 4i_0 + 6I_3 + 2I = 0$

$\Rightarrow -8I_2 + 9I_3 = -2$

KVL for Super Mesh (2,5):

$-5 + 3I_5 + 4i_0 + 4(I_2 - I_3) + 4I_2 = 0$

$\Rightarrow \frac{57}{5}I_2 - 8I_3 = 5$

$\Rightarrow I_2 = \frac{145}{193}A \approx 0.75A$

$I_3 = \frac{86}{193}A \approx 0.45A$

$i_0 = I_2 - I_3 \approx 0.31A$

$I_5 = -\frac{1}{5}I_2 = -\frac{29}{193}A$

$V_3 = (I_3 - I_4) \cdot 1\Omega \approx 2.45V$

$V_2 = 4\Omega \cdot i_0 = \frac{236}{193}V \approx 1.22V$

$V_1 = V_3 + v_0 = \frac{385}{193}V \approx 1.99V$